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Highlights

Sometimes you can't find the y -intercept by looking at the graph. The y -intercept may be a fraction, or it may intersect at a point not shown on the graph. In this lesson, you will learn to use the **point-slope form**. Use this form when you don't know the y -intercept.

The point-slope form appears on the GED math formula sheet.

point-slope form $y - y_1 = m(x - x_1)$, where m = slope and (x_1, y_1) is a point on the line

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Write the equation of the line passing through $(5, 2)$ and $(7, -2)$.

Step 1

Use the slope formula to find the slope. Let $(5, 2)$ be (x_1, y_1) and $(7, -2)$ be (x_2, y_2) .

$$m = \frac{-2-2}{7-5} = \frac{-4}{2} = -2$$

Step 2

Substitute the values into point-slope form. You may use either point.

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -2(x - 5)$$

Step 3

Rewrite the equation in slope-intercept form.

$$y - 2 = -2(x - 5)$$

$$y - 2 = -2x + 10$$

$$y = -2x + 12$$

The equation of the line is $y = -2x + 12$.

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Graph the equation $2x - 3y = 9$.

Step 1

Write the equation in slope-intercept form by isolating y .

$$2x - 3y = 9$$

$$-3y = -2x + 9$$

$$y = \frac{2}{3}x - 3$$

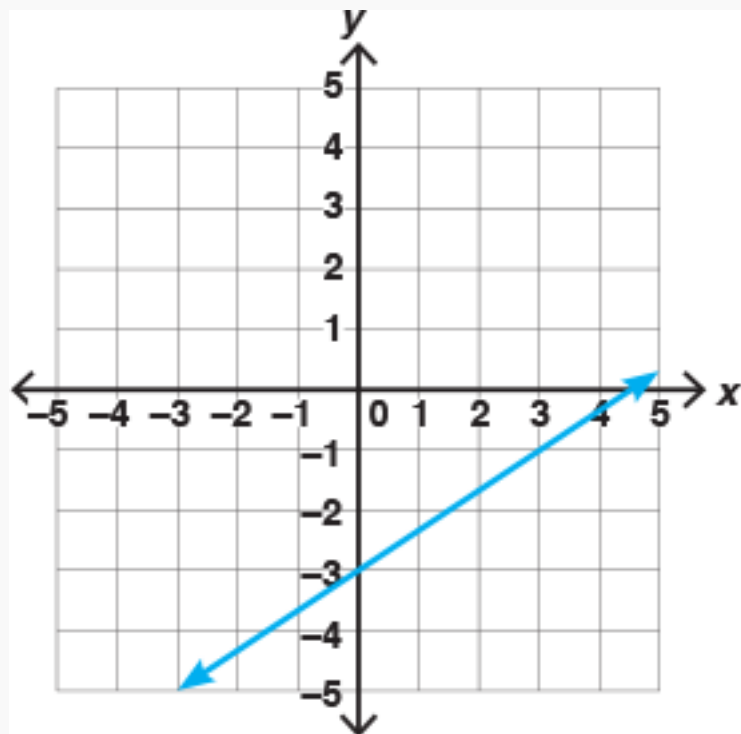
Step 2

Graph the equation using the slope and y -intercept.

$$y = \frac{2}{3}x - 3$$

The slope is $\frac{2}{3}$.

The y -intercept is $(0, -3)$.



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Category

Lesson 3: Writing Linear Equations

 **Related Questions:**
[Practice questions taken related to this topic](#)